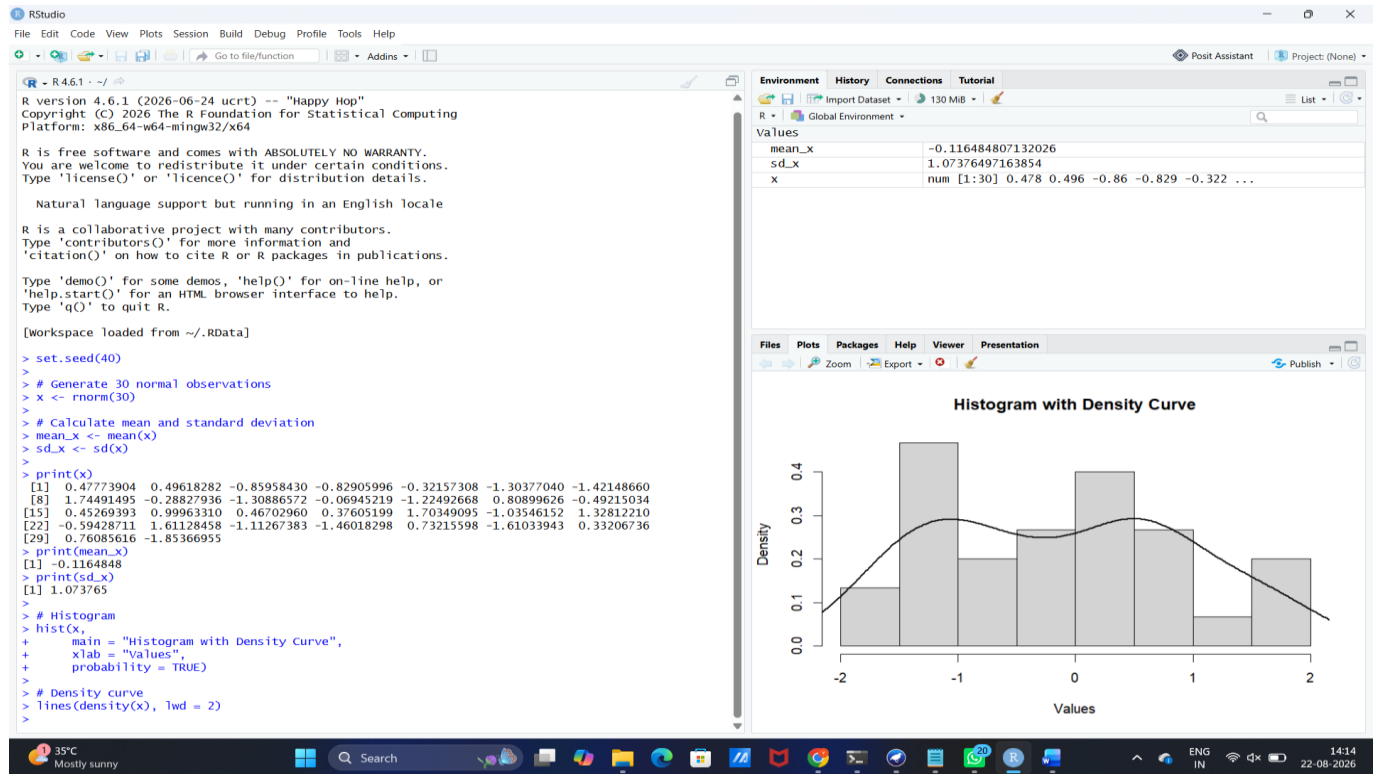
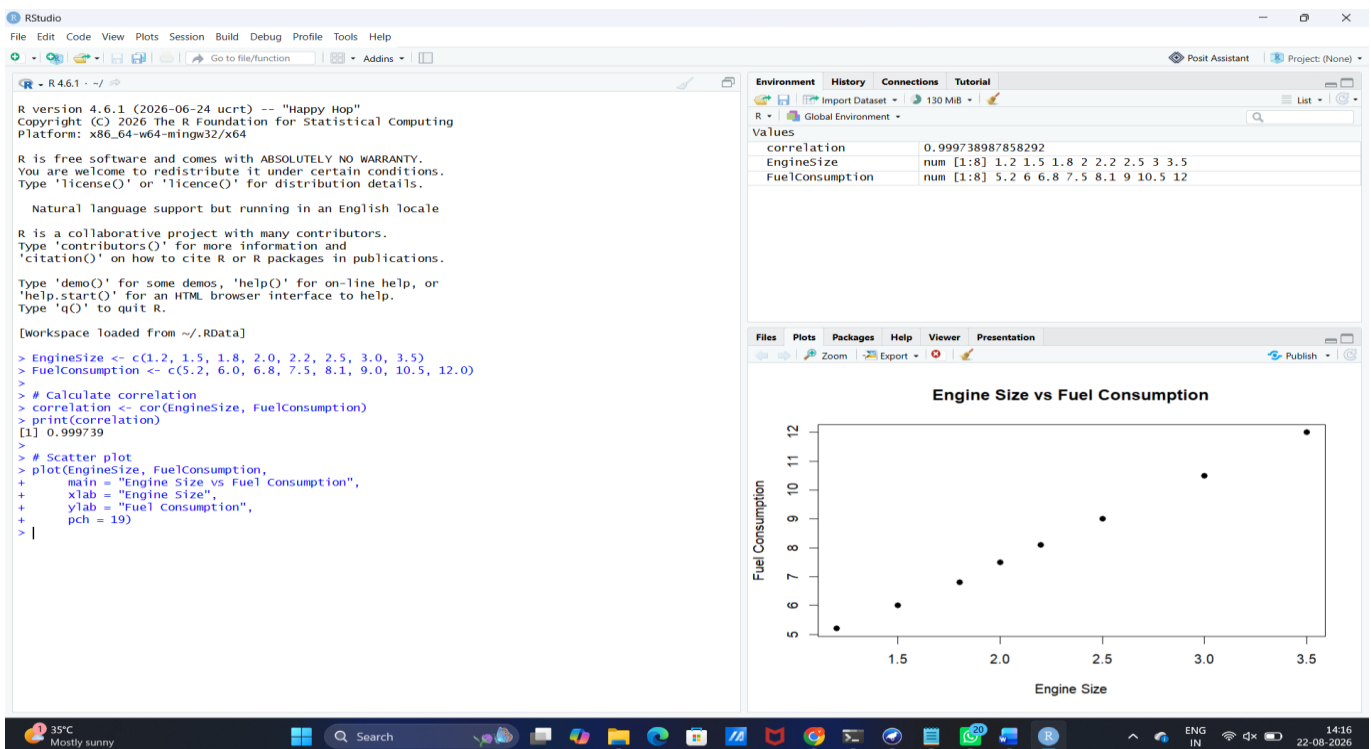


SET 14

1. R: Generate 30 normal observations with `set.seed(40)`; calculate mean and SD, draw histogram and density curve.



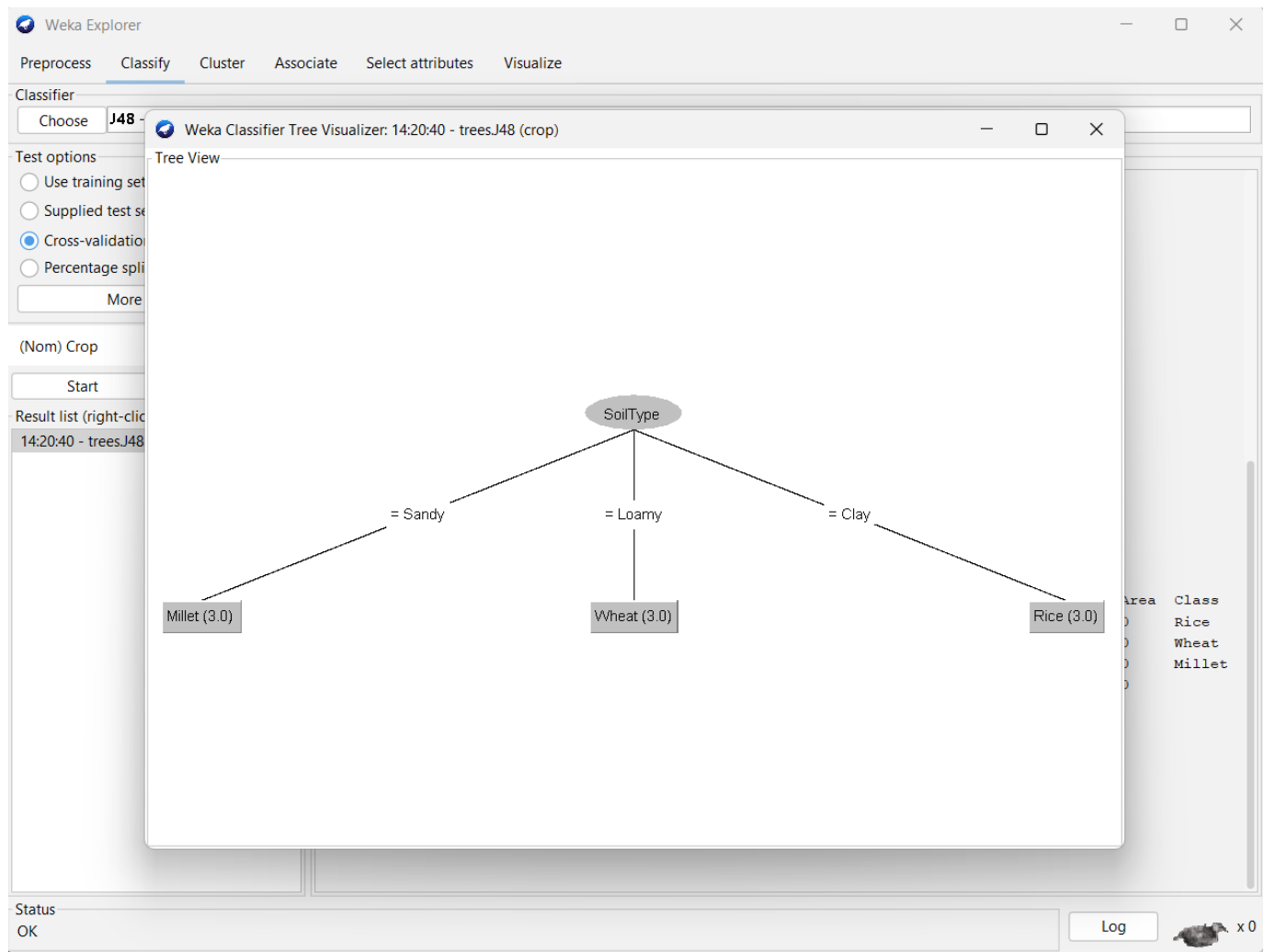
2. R: Create `EngineSize` and `FuelConsumption` vectors; draw scatter plot and calculate correlation.



3. WEKA: Create crop-classification ARFF data (SoilType, Rainfall, Temperature, Crop); apply J48 and visualize.

Dataset: Crop Classification

```
@relation crop
@attribute SoilType {Sandy,Loamy,Clay}
@attribute Rainfall numeric
@attribute Temperature numeric
@attribute Crop {Rice,Wheat,Millet}
@data
Clay,1200,28,Rice
Loamy,900,24,Wheat
Sandy,500,30,Millet
Clay,1100,27,Rice
Loamy,850,22,Wheat
Sandy,450,31,Millet
Clay,1300,29,Rice
Loamy,800,23,Wheat
Sandy,600,30,Millet
```



4. WEKA: Apply SMO to the crop data and compare precision, recall and F1 with J48.

Dataset: Crop Classification – SMO

Use the same Crop Classification ARFF dataset from Set 14 Q3.

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseSMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-

Test options

Use training set

Supplied test set

Cross-validation

Folds9

Percentage split

%66

More options...

(Nom) Crop

Start

Stop

Result list (right-click for options)

14:23:49 - functions.SMO

Classifier output

Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	9	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.2222		
Root mean squared error	0.2722		
Relative absolute error	45.8333	%	
Root relative squared error	52.9238	%	
Total Number of Instances	9		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Rice
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Wheat
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Millet
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a b c <-- classified as

3 0 0 | a = Rice

0 3 0 | b = Wheat

0 0 3 | c = Millet

Status

OK

Log

x 0